

ML Challenge 2025: Smart Product Pricing

Solution Report

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1. Executive Summary

We build a **rule-compliant** text+structured transformer regressor for product price prediction from `catalog_content`. Text is encoded with `distilbert-base-uncased` (frozen by default) while regex-derived quantity cues (*Pack of X*, *Value/Unit*, text ratios) form structured features. Embeddings are fused via a lightweight MLP to predict price with a log-space MSE objective and early stopping on SMAPE.

2. Methodology Overview

2.1 Problem Analysis

Supervised regression from catalog descriptions; quantity/packaging signals correlate strongly with price.

2.2 Approach

- **Text encoder:** DistilBERT tokenizer (max length configurable) $\rightarrow$ CLS embedding (encoder frozen unless `-fine_tune_text 1`).
- **Structured features:** From the same text using regex; normalized units, logs, and text statistics. Numeric scaling via `StandardScaler(with_mean=false)`; categorical `unit_base` one-hot with frequency cap (≥ 20 kept).
- **Fusion head:** Structured MLP + concatenation with text embedding $\rightarrow$ MLP $\rightarrow$ scalar log-price; final price $\hat{y} = \exp(z)$.
- **Training:** AdamW, cosine LR, mixed precision on CUDA, early stopping via validation SMAPE.

3. Model Architecture

3.1 Diagram

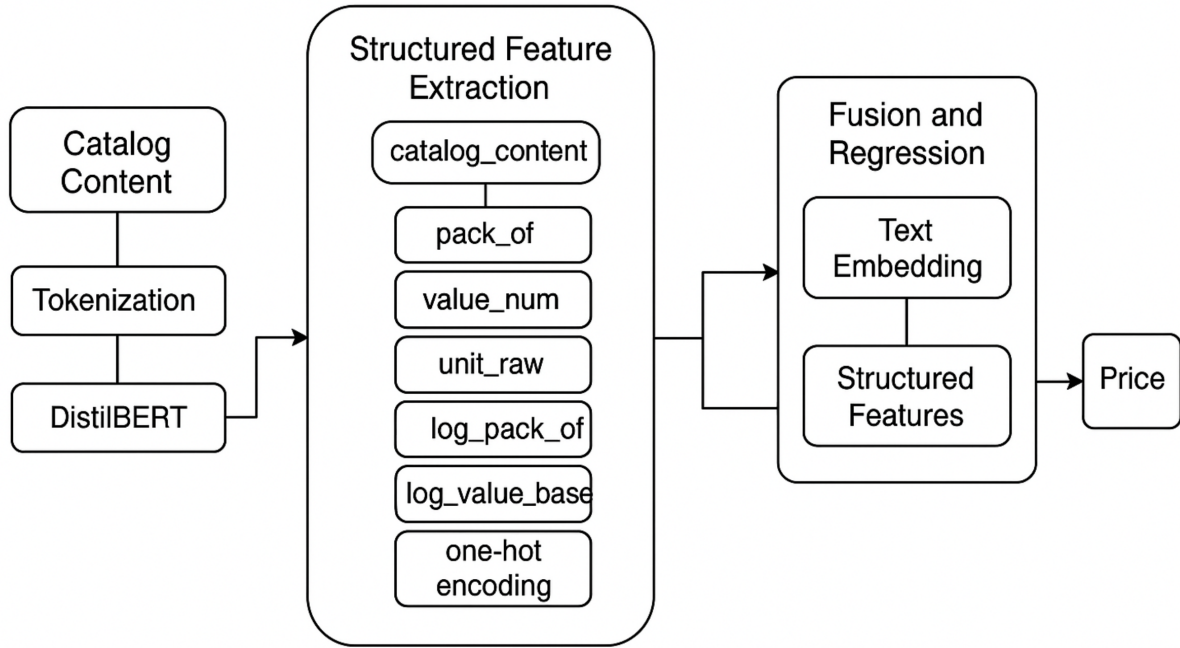


Figure 1: Pipeline — tokenizer + DistilBERT, structured feature builder, fusion MLP, price output.

3.2 Key Implementation Details

- Split via `train_test_split` with configurable `-val_size` (default 0.1) and `-seed`.
- Best checkpoint saved to `best_model_text.pth` and restored for test predictions.
- Robust clipping of predictions to $[0.01, Q_{0.997}]$ to reduce outlier impact.
- Submission written to the configured `-output_csv` (e.g., `./dataset/test_out.csv`).

4. Dataset, I/O and Compliance

4.1 Dataset Columns

- **sample_id** (unique ID), **catalog_content** (title+description+IPQ), **image_link** (optional), **price** (train only).

4.2 Output Format

CSV with exactly: `sample_id, price`. A prediction is produced for every `sample_id`.

4.3 Rule Compliance

- No external price lookup or internet augmentation.
- Only `catalog_content`-derived features are used in this submission.
- No third-party calls during training/inference; pretrained weights are standard NLP initialization.

5. Training Objective and Metric

Loss (training):

$$\mathcal{L} = \text{MSE}(\log(1 + \hat{y}), \log(1 + y)).$$

Model selection (validation): SMAPE:

$$\text{SMAPE} = 100 \times \frac{1}{N} \sum_i \frac{|\hat{y}_i - y_i|}{(|\hat{y}_i| + |y_i|)/2}.$$

6. Results

6.1 Training Log (this run)

Epoch	Time (s)	Train Loss	Val SMAPE (%)
1/5	1438.1	0.7594	62.846
2/5	1463.4	0.6831	64.874
3/5	1438.9	0.6567	62.149
4/5	1493.0	0.6263	60.404
5/5	1435.9	0.6126	60.014

*Average epoch time: ~ 1453.9 s. Best validation SMAPE: **60.014%**.*

6.2 Submission

Predictions were generated for the full test set and saved to `test_out.csv`.

7. Conclusion

A compact **transformer+structured** regressor—with engineered quantity features and a frozen DistilBERT backbone—achieves a best validation SMAPE of **60.014%** on this run while complying with all challenge rules.